Individual Project (CS3IP)

Department of Computer Science
University of Reading

Project Initiation Document

PID Sign-Off

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Degree programme (BSc CS/BSc CSwIY)	**Computer Science**
Supervisor Name *(Consultation with supervisor is mandatory)*	**Pat parslow**
	Supervisor to sign PID form on Bb (grade centre)
Date	

SECTION 1 – General Information

Project Identification

1.1	**Project Title**
	Pyschological Narrative Engine (PNE)
1.2	**Please describe the project with key-phrases (max 5)**
	Systems-Thinking, Emergent Narrative, Dynamic Social Simulation, Cognitive Modelling
1.3	**E-logbook maintenance agreed with supervisor** *Use Google doc, OneDrive, or any mobile App whereby you will be able to generate a PDF copy*
	Google Doc
1.4	**GitLab link for maintain source code and research data** *Any change in GitLab link and Source code repository MUST be explicitly mention in final report*

SECTION 2 – Project Description

2.1	**Summarise the project's background in terms of research field /application domain (max 100 words).**
	This project explores the intersection of computer science, systems thinking and psychology by designing a psychological narrative engine (PNE.) Using a game as the interface as a simulation medium. The PNE models how internal cognition, social interactions and observer influences dynamically affect emergent narrative outcomes. By combining different social relationship systems, public image systems and feedback loops, the project aims to bridge cognitive psychology with gameplay to demonstrate how player perception and decision-making can be algorithmically represented and tested within interactive digital environments
2.2	**Summarise the project aims, objectives and outputs (max 250 words).** These aims, objectives, and outputs should appear as the tasks, milestones and deliverables in your project plan (fill out Section 3).
	- To design and prototype a Psychological Narrative Engine that simulates an emergent narrative based on player psychology and perception - To demonstrate how system thinking principles can model cognitive and social interactions in a computational environment. Objectives - Define Core PNE entities. The Internal (The Player's own reflective / introspection), The External (The NPCS at the environment) and an interface (Player selected Dialouge) - Develop a framework to manage player abilities and social standing - Document system architecture, interaction models and evaluation of emergent behaviour within a CS framework. Outputs - Working prototype demonstrating PNE System applied to gameplay mechanics - System architecture diagrams and data structures bridging CS and psychologu - Working game showing an emergent narrative
2.3	**Initial project specification – roughly indicate key features and functions of your finished program/application. Indicate possible method, data source, technology etc. (max 400 words)** (Sensible and relevant Charts, Table, and Figures can be used)

	- Internal Perception System: Generates internal monologue and thoughts influenced by the Player's action and social positioning Tracks cognitive alignment and bias to produce emergent narrative interpretations Dynamic NPC System: NPC's are somewhat autonomous agents. They have emotional relational and archetypal traits Interactions respond to player interal state, actions, and feedback loops demonstrating adaptive behaviour.
2.4	**Describe the social, legal and ethical issues that apply to your project. Does your project require ethical approval? (If your project requires a questionnaire/interview for conducting research and/or collecting data, you will need to apply for an ethical approval)**
	- No personal data collection - Psychological constructs are abstract and symbolic / merely suggestions rather than diagnosis - Compliance with software licences / copyright procedures.
2.5	**Identify the items you may need to purchase for your project. A cost upto £200 can be applied (include VAT and shipping if known). You need to have consent of your supervisor. Your request will be assessed by the department.**
2.6	**State whether you need access to specific resources within the department or the University e.g. special devices and workshop**
	- Lab PCs / high-spec personal workstation

SECTION 3 - Project Plan

Task / Week	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35
Background Research																																				
Literature review (systems thinking, cognitive modeling, procedural narrative)	▶																																			
Survey existing psychological narrative engines & social simulations			▶																																	
Research AI dialogue systems & branching structures					▶																															
Analysis / Design																																				
Define core PNE entities (Internal Reflection, External Social System)							▶																													
Design social & skill framework (Empathy, Novelty, Combat, Language)									▶																											
Map feedback loops, dynamic NPC behaviors, integrity metrics												▶																								
Document system architecture & interaction models														▶																						
Prototype Development																																				
Implement Internal Perception system																▶			◀																	
Implement Dynamic NPC system with adaptive responses																			▶				◀													
Integrate skill influence into monologue, dialogue & outcomes																							▶		◀											
Create minimal test scenarios for emergent narratives																									1w	1w										
Testing / Evaluation / Validation																																				
Unit testing of internal systems & skill mechanics																									▶	◀										
System integration testing (PNE + NPCs + Skills)																										▶		◀								
Validation of emergent narrative outcomes																												▶								
Iterative refinement based on feedback																														1w						
Assessment / Documentation																																				
Compile architecture diagrams, data structures & models																															▶	◀				
Document design rationale, emergent behavior & implementation																																▶		◀		
Write final report (CS & psychology integration)																																		▶		◀
Submit prototype, report & deliverables																																				0.5w